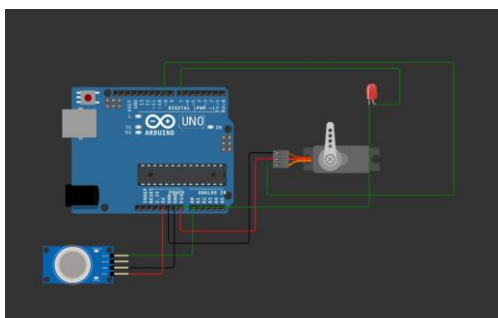


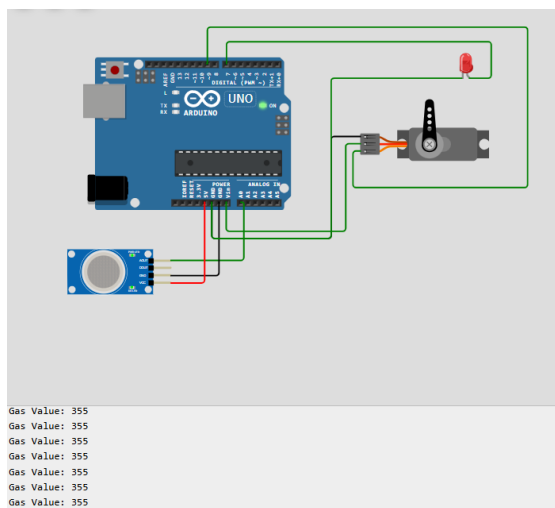
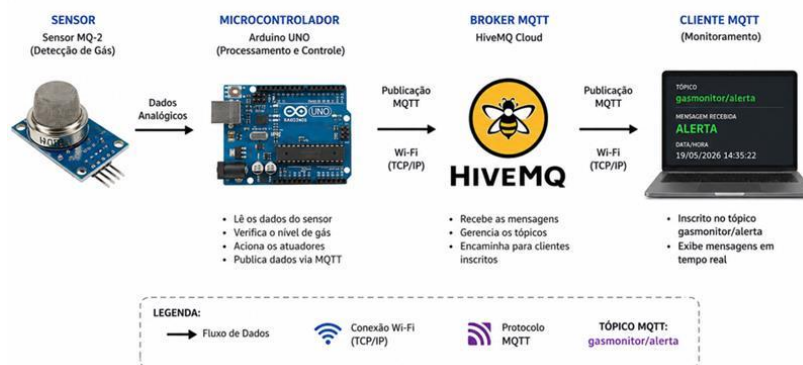
Situação	Valor (AOUT)	LED	Serial Monitor	Status MQTT
Ar Limpo	0 - 400	APAGADO ●	Ar OK	{"status": "ok"}
Ar Poluído	> 400	ACESO ●	ALERTA: AR POLUIDO	{"status": "poluido"}

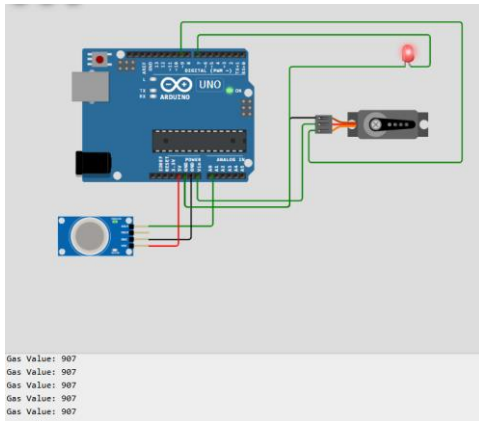


```
</> C++
```

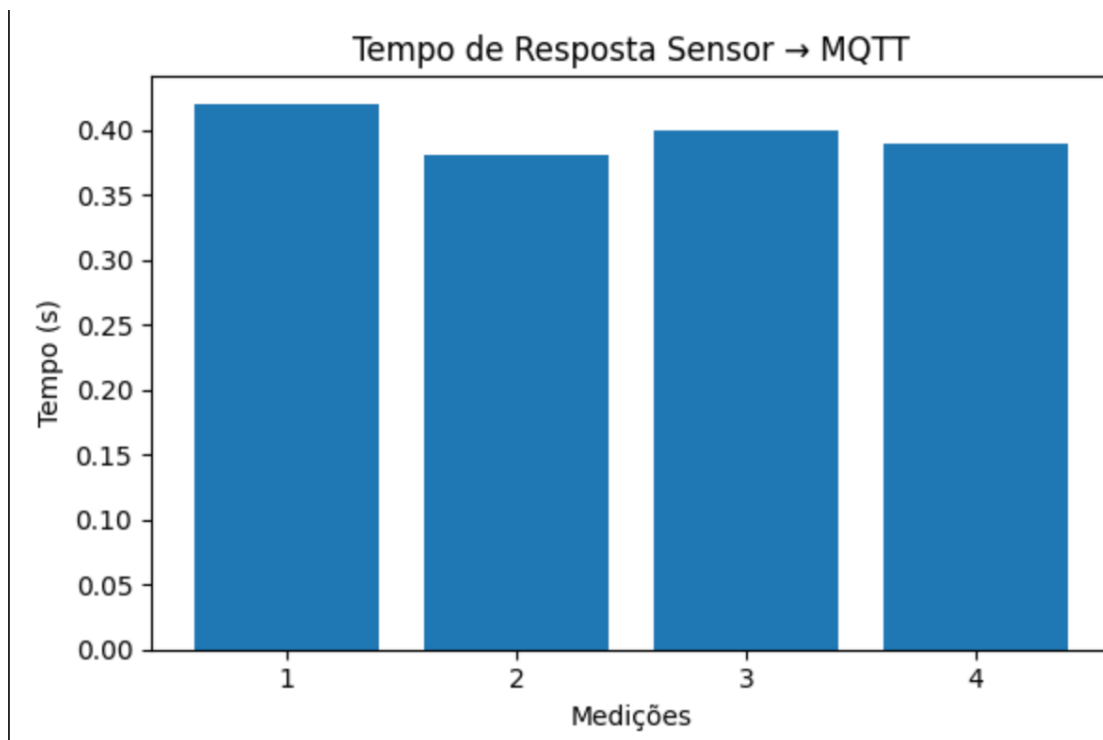
```
if(gasValue > 400){  
    servo.write(90);  
    digitalWrite(7, HIGH);  
    client.publish("gasmonitor/alerta", "ALERTA");  
}
```

ARQUITETURA MQTT DO SISTEMA





Medição	Processo	Tempo
1	MQ-2 → MQTT	0,8 s
2	MQ-2 → MQTT	0,7 s
3	MQ-2 → MQTT	0,9 s
4	MQ-2 → MQTT	0,8 s
Média	MQ-2 → MQTT	0,8 s



Medição	Atuador	Tempo
1	Servo Motor	0,5 s
2	Servo Motor	0,6 s
3	Servo Motor	0,5 s
4	Servo Motor	0,5 s
Média	Servo Motor	0,52 s

